

**MARK SCHEME for the October/November 2009 question paper
for the guidance of teachers**

9709 MATHEMATICS

9709/22

Paper 22, maximum raw mark 50

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes must be read in conjunction with the question papers and the report on the examination.

- CIE will not enter into discussions or correspondence in connection with these mark schemes.

CIE is publishing the mark schemes for the October/November 2009 question papers for most IGCSE, GCE Advanced Level and Advanced Subsidiary Level syllabuses and some Ordinary Level syllabuses.



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Mark Scheme Notes

Marks are of the following three types:

M Method mark, awarded for a valid method applied to the problem. Method marks are not lost for numerical errors, algebraic slips or errors in units. However, it is not usually sufficient for a candidate just to indicate an intention of using some method or just to quote a formula; the formula or idea must be applied to the specific problem in hand, e.g. by substituting the relevant quantities into the formula. Correct application of a formula without the formula being quoted obviously earns the M mark and in some cases an M mark can be implied from a correct answer.

A Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. Accuracy marks cannot be given unless the associated method mark is earned (or implied).

B Mark for a correct result or statement independent of method marks.

- When a part of a question has two or more "method" steps, the M marks are generally independent unless the scheme specifically says otherwise; and similarly when there are several B marks allocated. The notation DM or DB (or dep*) is used to indicate that a particular M or B mark is dependent on an earlier M or B (asterisked) mark in the scheme. When two or more steps are run together by the candidate, the earlier marks are implied and full credit is given.
- The symbol $\checkmark$ implies that the A or B mark indicated is allowed for work correctly following on from previously incorrect results. Otherwise, A or B marks are given for correct work only. A and B marks are not given for fortuitously "correct" answers or results obtained from incorrect working.
- Note: B2 or A2 means that the candidate can earn 2 or 0.
B2/1/0 means that the candidate can earn anything from 0 to 2.

The marks indicated in the scheme may not be subdivided. If there is genuine doubt whether a candidate has earned a mark, allow the candidate the benefit of the doubt. Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored.

- Wrong or missing units in an answer should not lead to the loss of a mark unless the scheme specifically indicates otherwise.
- For a numerical answer, allow the A or B mark if a value is obtained which is correct to 3 s.f., or which would be correct to 3 s.f. if rounded (1 d.p. in the case of an angle). As stated above, an A or B mark is not given if a correct numerical answer arises fortuitously from incorrect working. For Mechanics questions, allow A or B marks for correct answers which arise from taking g equal to 9.8 or 9.81 instead of 10.

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The following abbreviations may be used in a mark scheme or used on the scripts:

AEF	Any Equivalent Form (of answer is equally acceptable)
AG	Answer Given on the question paper (so extra checking is needed to ensure that the detailed working leading to the result is valid)
BOD	Benefit of Doubt (allowed when the validity of a solution may not be absolutely clear)
CAO	Correct Answer Only (emphasising that no "follow through" from a previous error is allowed)
CWO	Correct Working Only - often written by a 'fortuitous' answer
ISW	Ignore Subsequent Working
MR	Misread
PA	Premature Approximation (resulting in basically correct work that is insufficiently accurate)
SOS	See Other Solution (the candidate makes a better attempt at the same question)
SR	Special Ruling (detailing the mark to be given for a specific wrong solution, or a case where some standard marking practice is to be varied in the light of a particular circumstance)

Penalties

MR -1	A penalty of MR -1 is deducted from A or B marks when the data of a question or part question are genuinely misread and the object and difficulty of the question remain unaltered. In this case all A and B marks then become "follow through ✓" marks. MR is not applied when the candidate misreads his own figures - this is regarded as an error in accuracy. An MR-2 penalty may be applied in particular cases if agreed at the coordination meeting.
PA -1	This is deducted from A or B marks in the case of premature approximation. The PA -1 penalty is usually discussed at the meeting.

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- 1 *EITHER:* Obtain a non-modular inequality from $(x + 3)^2 > (2x)^2$, or corresponding equation, or pair of linear equations $(x + 3) = \pm 2x$ M1
 Make reasonable solution attempt at a 3-term quadratic, or solve two linear equations M1
 Obtain critical values $x = -1$ and $x = 3$ A1
 State answer $-1 < x < 3$ A1
OR: Obtain critical value $x = 3$ from a graphical method, or by inspection, or by solving a linear inequality or linear equation B1
 Obtain the critical value $x = -1$ similarly B2
 State answer $-1 < x < 3$ B1 [4]
- 2 State or imply $2 \ln x = \ln(x^2)$ B1
 Use law for the logarithm of a quotient or a product M1
 Remove logarithms and obtain $yx^2 = y + 5$, or equivalent A1
 Obtain answer $y = \frac{5}{x^2 - 1}$ A1 [4]
- 3 (i) Show or imply correct ordinates 1, 1.15470..., 2 B1
 Use correct formula, or equivalent, with $h = \frac{1}{6}\pi$ and three ordinates M1
 Obtain answer 1.39 with no errors seen A1 [3]
- (ii) Make recognisable sketch of $y = \sec x$ for $0 \leq x \leq \frac{1}{3}\pi$ B1
 Using a correct graph, explain that the rule gives an over-estimate B1 [2]
- 4 (i) State $\frac{dx}{dt} = e^{-t}$ or $\frac{dy}{dt} = e^t - e^{-t}$ B1
 Use $\frac{dy}{dx} = \frac{dy}{dt} \div \frac{dx}{dt}$ M1
 Obtain given answer correctly A1 [3]
- (ii) Substitute $\frac{dy}{dx} = 2$ and use correct method for solving an equation of the form $e^{2t} = a$, where $a > 0$ M1
 Obtain answer $t = \frac{1}{2} \ln 3$, or equivalent A1 [2]
- 5 (i) Substitute $x = -1$ or $x = 2$ and equate to zero M1
 Obtain a correct equation, e.g. $-a + b + 5 + 2 = 0$ A1
 Obtain a second correct equation, e.g. $8a + 4b - 10 + 2 = 0$ A1
 Solve for a or b M1
 Obtain $a = 3$ and $b = -4$ A1 [5]
- (ii) Substitute for a and b and attempt division by $(x + 1)(x - 2)$ or attempt third factor by inspection M1
 Obtain answer $3x - 1$ A1 [2]

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6	(i)	State answer $R = 5$ Use trig formula to find a Obtain $a = 53.13^\circ$	B1 M1 A1	[3]
	(ii)	Evaluate $\cos^{-1}(4.5/5) \approx 25.84^\circ$ Obtain answer 79.0° Carry out correct method for second answer Obtain answer 27.3° and no others in the given range [Treat the giving of answers in radians as a misread. Ignore answers outside the given range.]	M1 A1 M1 A1✓	
7	(i)	Use product rule Obtain correct derivative in any form Equate derivative to zero and express $\tan x$ in terms of x Obtain given answer	M1 A1 M1 A1	[4]
	(ii)	Consider sign of $\tan x - \frac{2}{x}$ at $x = 1$ and $x = 1.2$, or equivalent Complete the argument with correct calculations	M1 A1	
	(iii)	Use the iterative formula correctly at least once Obtain final answer 1.08 Show sufficient iterations to justify its accuracy to 2 d.p. or show there is a sign change in the interval (1.075, 1.085)	M1 A1 A1	[3]
	(a)	Integrate and obtain term $k \cos 2x$, where $k = \pm \frac{1}{2}$ or ± 1 Obtain term $-\frac{1}{2} \cos 2x$ Obtain term $\tan x$ Substitute correct limits correctly Obtain exact answer $\frac{3}{4} + \sqrt{3}$	M1 A1 B1 M1 A1	
8	(b)	Integrate and obtain $\frac{1}{2} \ln x + \ln(x + 1)$ or $\frac{1}{2} \ln 2x + \ln(x + 1)$ Substitute correct limits correctly Obtain given answer following full and correct working	B1 + B1 M1 A1	[4]